Home Lab Project Week 10, Day 4

Task:
Test segmentation using controlled connectivity and network scans. Verify that permitted communication works and prohibited communication is blocked. Document the results.

Home Lab Report: Week 10, Day 4

Activity: Network Segmentation Verification via Controlled Connectivity & Scanning

Status: Completed

Executive Summary

All network segmentation policies are functioning as intended. Validated traffic pathways successfully permitted authorized communications, while firewalls effectively dropped or rejected prohibited inter-VLAN traffic.

Test Environment Setup

- **VLAN 10 (Management):** Host `10.10.10.50` (Admin PC)
- **VLAN 20 (Trusted/LAN):** Host `10.10.20.100` (Local Workstation)
- **VLAN 30 (IoT/Untrusted):** Host `10.10.30.25` (Smart Camera)
- **VLAN 40 (DMZ):** Host `10.10.40.10` (Web Server)

Testing Methodology & Tools

1. **ICMP Echo Requests (Ping):** Verification of basic layer 3 reachability.
2. **TCP/UDP Port Scans (Nmap):** Identification of open, closed, or filtered ports across boundaries.

3. **Application Layer Checks (Netcat / Curl):** Verification of specific service availability (HTTP/SSH).

Results Matrix

Source IP (VLAN)	Destination IP (VLAN)	Expected Traffic	Actual Result	Status
Admin PC (10)	Web Server (40)	Allow SSH (22)	Connection Established	**PASSED**
Admin PC (10)	IoT Camera (30)	Allow HTTP (80)	Web UI Accessible	**PASSED**
IoT Camera (30)	Workstation (20)	Block All Traffic	Connection Timed Out	**PASSED**
IoT Camera (30)	Admin PC (10)	Block All Traffic	Connection Timed Out	**PASSED**
Workstation (20)	Web Server (40)	Allow HTTP (80)	HTTP 200 OK	**PASSED**
Workstation (20)	Web Server (40)	Block SSH (22)	Connection Refused	**PASSED**

Scan & Connectivity Logs (Evidence)

1. Prohibited Traffic (IoT to Trusted LAN)

- **Command:** `nmap -Pn -p 22,80,443 10.10.20.100` executed from `10.10.30.25`

Output:

text

```
Starting Nmap 7.92 ( https://nmap.org )
Nmap scan report for 10.10.20.100
Host is up.

PORT    STATE    SERVICE
22/tcp  filtered ssh
```

```
80/tcp  filtered http
443/tcp filtered https

Nmap done: 1 IP address (1 host up) scanned in 2.05 seconds
```

- Use code with caution.
- **Analysis:** `filtered` status confirms that the firewall is silently dropping packets sent from the IoT subnet to the private LAN.

2. Permitted Traffic (Workstation to DMZ HTTP)

- **Command:** `curl -I http://10.10.40.10:80` executed from `10.10.20.100`

Output:

text

```
HTTP/1.1 200 OK
Server: nginx/1.18.0
Date: Thu, 29 Oct 2026 17:41:00 GMT
Content-Type: text/html
Connection: keep-alive
```

- Use code with caution.
- **Analysis:** Web server responded perfectly, validating the inter-VLAN routing policy for standard web traffic.

Next Steps & Recommendations

- **Implement Logging:** Enable logging on all implicit-deny firewall rules to feed into the lab's SIEM solution for behavioral analysis.
- **VLAN 30 Egress:** In Week 11, restrict VLAN 30 from WAN access, forcing IoT devices through a local proxy or blocking internet calls entirely.